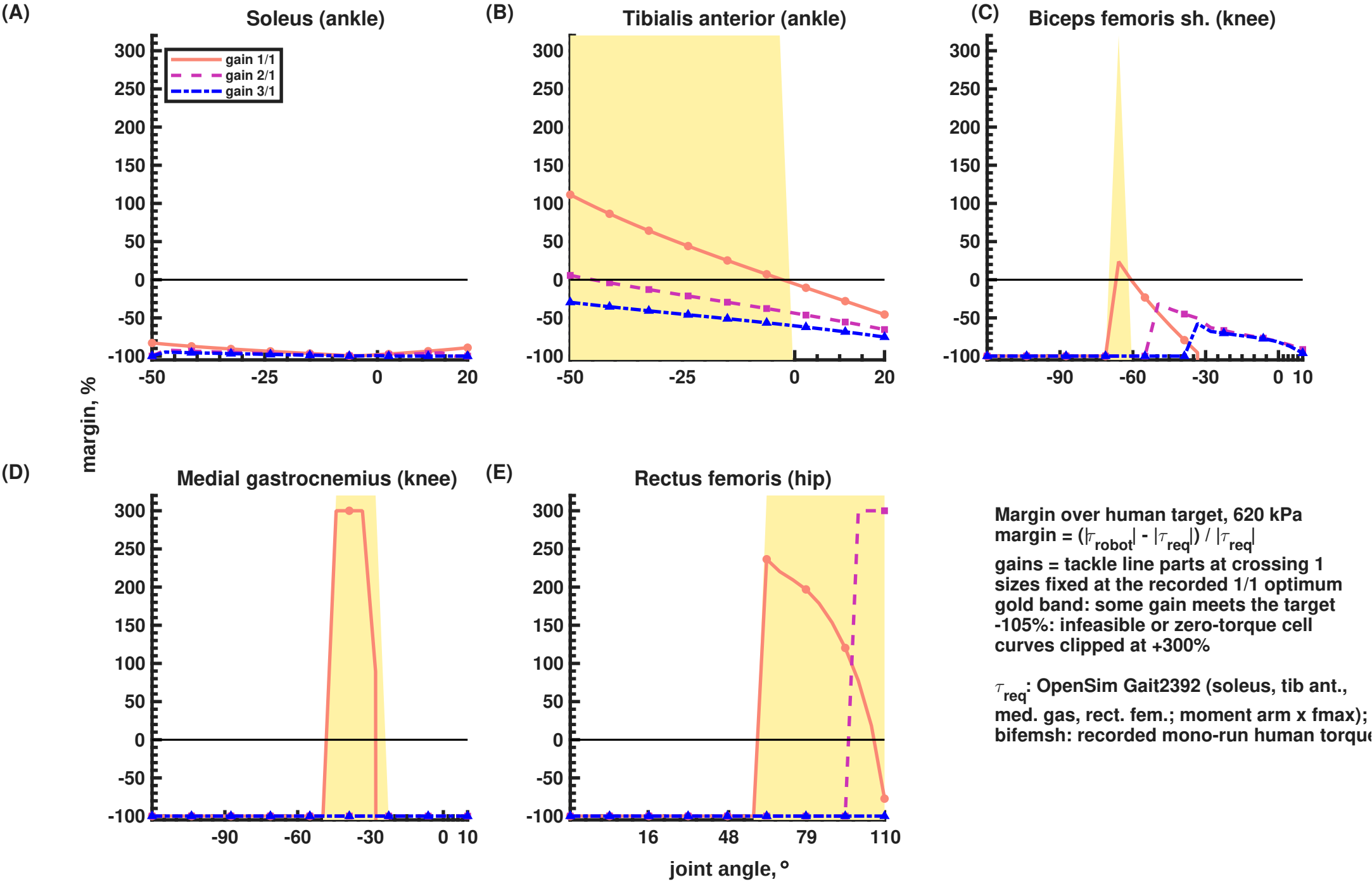
(A)
Soleus (ankle)
gain 1/1
gain 2/1
gain 3/1
(B)
Tibialis anterior (ankle)
(C)
Biceps femoris sh. (knee)
margin, %
(D)
Medial gastrocnemius (knee)
(E)
Rectus femoris (hip)
joint angle, °
Margin over human target, 620 kPa
margin = (|τrobot| - |τreq|) / |τreq|
gains = tackle line parts at crossing 1
sizes fixed at the recorded 1/1 optimum
gold band: some gain meets the target
-105%: infeasible or zero-torque cell
curves clipped at +300%
τreq: OpenSim Gait2392 (soleus, tib ant.,
med. gas, rect. fem.; moment arm x fmax);
bifemsh: recorded mono-run human torque